

50.039 Theory and Practice of Deep Learning

W1-S2 Introduction and Machine Learning Reminders

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About this week (Week 1)

1. What are the **typical concepts of Machine Learning** to be used as a starting point for this course?
2. What are the **different families of problems** in Deep Learning?
3. What is the **typical structure of a Deep Learning problem**?
4. What is **linear regression** and how to implement it?
5. What is the **gradient descent algorithm** and how is it used to **train Machine Learning models**?
6. What is **polynomial regression** and how to implement it?
7. What is **regularization** and how to implement it in **Ridge regression**?

About this week (Week 1)

8. What is **overfitting** and why is it bad?
9. What is **underfitting** and why is it bad?
10. What is **generalization** and how to evaluate it?
11. What is a **train-test split** and why is it related to **generalization**?
12. What is a **sigmoid** function? What is a **logistic** function?
13. How to perform **binary classification** using a **logistic regressor** and how is it related to linear regression?

About this week (Week 1)

14. What are **Neural Networks** and how do they relate to the **biology of a human brain**?
15. What is a **Neuron** in a Neural Network and how does it relate to linear/logistic regression?
16. What is the **difference** between a **shallow** and a **deep neural network**?
17. How to **implement a shallow Neural Network** manually and define a **forward propagation** method for it?
18. How to **train a shallow Neural Network** using **backpropagation**?
How to define **backward propagation** and **trainer** functions?

Extension of multi-parameter Linear Regression

Let us now consider that the inputs consist of more than one parameter, e.g. each sample x_i consists of:

- The surface in sqm x_i^1 ,
- The distance to the closest MRT station in meters x_i^2 ,

The linear regression can be simply transposed from a single input to multiple inputs.

s\$ 1,680,000

Negotiable

3  3  1184 sqft s\$ 1,418.92 psf

Details

Property Type

Executive Condominium For Sale

Floor Size

1184 sqft

Developer

Tampines EC Pte Ltd

PSF

S\$ 1,418.92 psf

Furnishing

Partially Furnished

Floor Level

High Floor

Tenure

99-year Leasehold

TOP

January, 2016



Extension of multi-parameter Linear Regression

Definition (Multi-parameter Linear Regression):

Multi-parameter Linear Regression is a model, which **assumes that there is a linear relationship between inputs and outputs.**

It therefore consists of several trainable parameters $(a_1, a_2, \dots, a_K, b)$, to be freely chosen.

These will connect any input $x_i = (x_i^1, x_i^2, \dots, x_i^K)$ to its respective output y_i , with the following equation:

$$y_i \approx \sum_{k=1}^K a_k x_i^k + b$$

Extension of multi-parameter Linear Regression

The MSE loss then becomes:

$$L = \frac{1}{N} \sum_{i=1}^N \left(\sum_{k=1}^K a_k x_i^k + b - y_i \right)^2$$

And the optimization problem related to training is then:

$$(a_1^*, \dots, a_K^*, b^*) = \arg \min_{a_1, \dots, a_K, b} L$$

Extension of multi-parameter Linear Regression

We could then try to compute the new normal equation for this problem (good practice for your optimization skills, try it out!)

Or, and it is often preferable, we could then define gradient descent update rules with respect to every trainable parameter $(a_1, a_2, \dots, a_K, b)$ like before.

However, we leave this implementation for students who would like to practice. (Or it might be the homework for a later week!).

From Linear to Polynomial Regression

Definition (Polynomial Regression with degree K):

Polynomial Regression is a model, which **assumes that there is a polynomial relationship between inputs and outputs**, which can be defined as a **polynomial function of degree K** .

It consists of several trainable parameters $(a_1, a_2, \dots, a_K, b)$, to be freely chosen.

These will connect any input x_i (e.g. the surface only) to its respective output y_i , with the same sort of polynomial equation with degree K :

$$y_i \approx \sum_{k=1}^K a_k (x_i)^k + b$$

From Linear to Polynomial Regression

We will follow the same steps as before with linear regression, starting with a mock dataset.

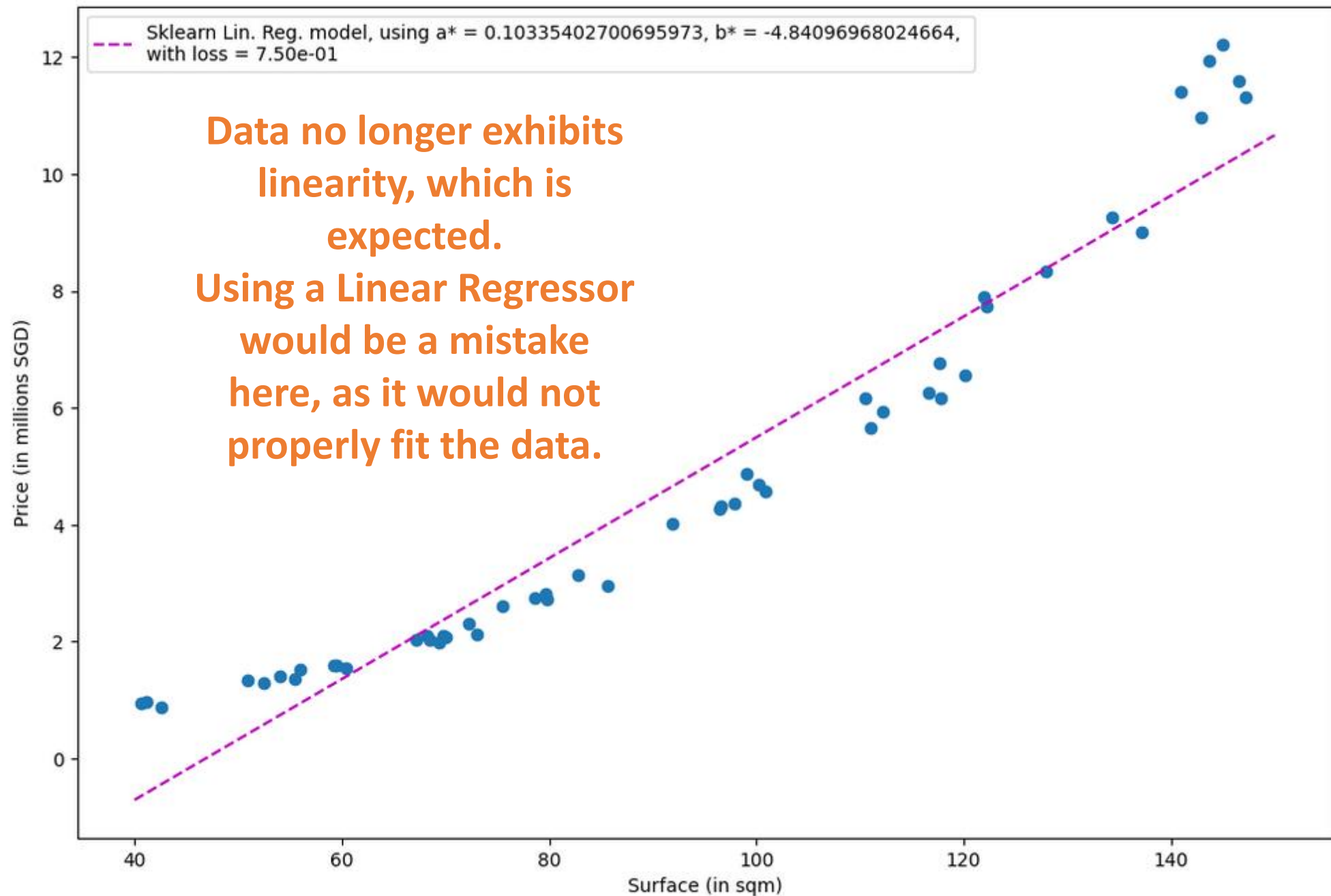
- This will be different compared than what has been implemented in notebooks 1 and 2, as we will generate prices y_i as a polynomial function of the surfaces x_i .
- We will assume that the function $y = f(x)$, giving the price of an apartment with surface x , is defined as a polynomial function with degree 3.

$$y = f(x) = 100000 + 14373x + 3x^3$$

- In addition, we will add a random noise to the final pricing, with a random +/- 10% drift as before.

From Linear to Polynomial Regression

```
1 # All helper functions
2 def surface(min_surf, max_surf):
3     return round(np.random.uniform(min_surf, max_surf), 2)
4 def price(surface):
5     # Note: this has changed and is now a polynomial function.
6     return round((100000 + 14373*surface + 3*surface**3)*(1 + np.random.uniform(-0.1, 0.1)))/1000000
7 def generate_datasets(n_points, min_surf, max_surf):
8     x = np.array([surface(min_surf, max_surf) for _ in range(n_points)])
9     y = np.array([price(i) for i in x])
10    return x, y
11 def linreg_matplotlib(a, b, min_surf, max_surf, n_points = 50):
12     x = np.linspace(min_surf, max_surf, n_points)
13     y = a*x + b
14     return x, y
15 def loss_mse(a, b, x, y):
16     val = np.sum((y - (a*x + b))**2)/x.shape[0]
17     return '{:.2e}'.format(val)
```



Polynomial Regression with sklearn

Sklearn treats polynomial regression as a **multi-parameter linear regression**, with **polynomial features**.

Polynomial features: reworking the inputs so that

$$\tilde{X} = \begin{pmatrix} x_1 & \cdots & (x_1)^K \\ \vdots & \ddots & \vdots \\ x_N & \cdots & (x_N)^K \end{pmatrix}$$

And $\tilde{x}_{i,j} = (x_i)^j$

```
1 # Preparing polynomial features for our dataset
2 n_degree = 3
3 sk_poly = PolynomialFeatures(degree = n_degree, include_bias = False)
4 sk_poly_inputs = sk_poly.fit_transform(sk_inputs.reshape(-1, 1))
5 print(sk_poly_inputs)
```

```
[5.24800000e+01 2.75415040e+03 1.44537813e+05]
[1.47190000e+02 2.16648961e+04 3.18885606e+06]
[1.20160000e+02 1.44384256e+04 1.73492122e+06]
[7.86600000e+01 6.18739560e+03 4.86700538e+05]
[1.17840000e+02 1.38862656e+04 1.63635754e+06]
[1.27960000e+02 1.63737616e+04 2.09518653e+06]
[1.11010000e+02 1.23232201e+04 1.36800066e+06]
[8.56100000e+01 7.32907210e+03 6.27441862e+05]
[1.17660000e+02 1.38438756e+04 1.62887040e+06]
[6.71300000e+01 4.50643690e+03 3.02517109e+05]
[6.81600000e+01 4.64578560e+03 3.16656746e+05]
[4.26400000e+01 1.81816960e+03 7.75267517e+04]
[5.08600000e+01 2.58673960e+03 1.31561576e+05]
[7.30500000e+01 5.33630250e+03 3.89816898e+05]
[1.10490000e+02 1.22080401e+04 1.34886635e+06]
[7.54400000e+01 5.69119360e+03 4.29343645e+05]
[6.04000000e+01 3.64816000e+03 2.20348864e+05]
[1.40000000e+02 1.00400000e+04 2.70666520e+06]
```

Polynomial Regression with sklearn

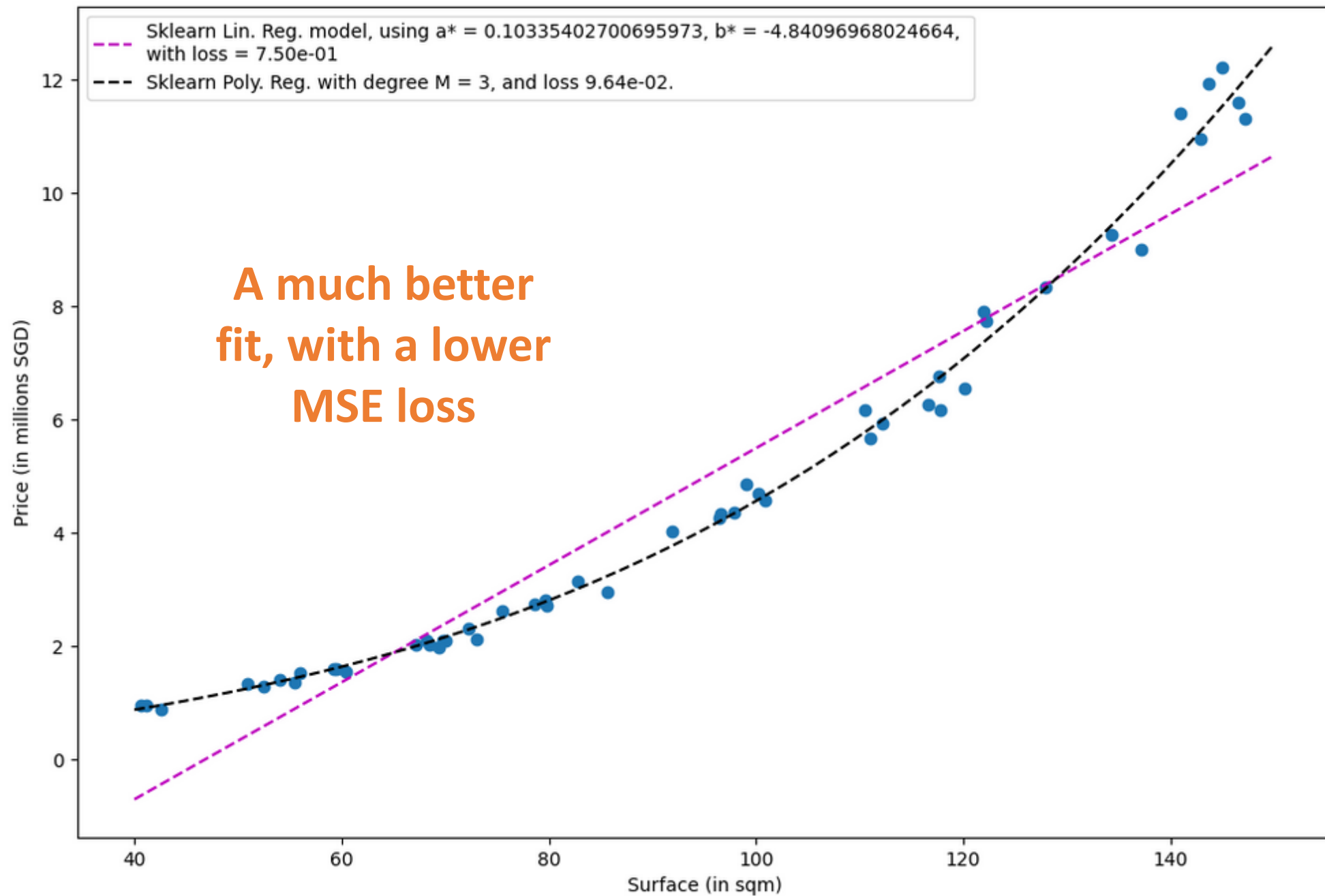
We then proceed normally, assuming $\tilde{X}$ are our new inputs.

Sklearn then automatically adjusts and produces the right number of parameters for the model.

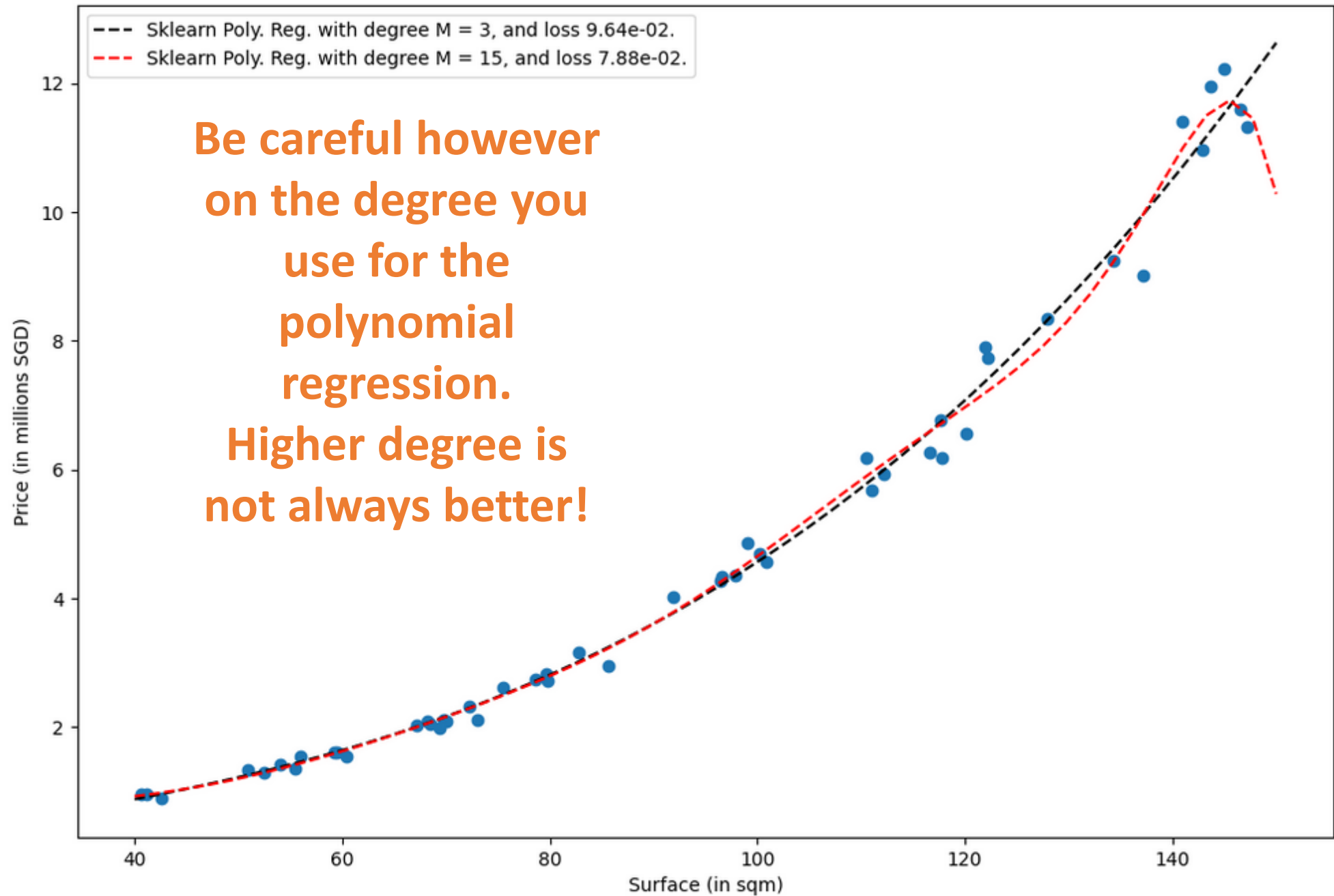
```
1 # Training a Polynomial Regressor
2 poly_reg_model = LinearRegression()
3 poly_reg_model.fit(sk_poly_inputs, sk_outputs)
4 a_sk_poly = poly_reg_model.coef_
5 b_sk_poly = poly_reg_model.intercept_
6 print(a_sk_poly, b_sk_poly)
```

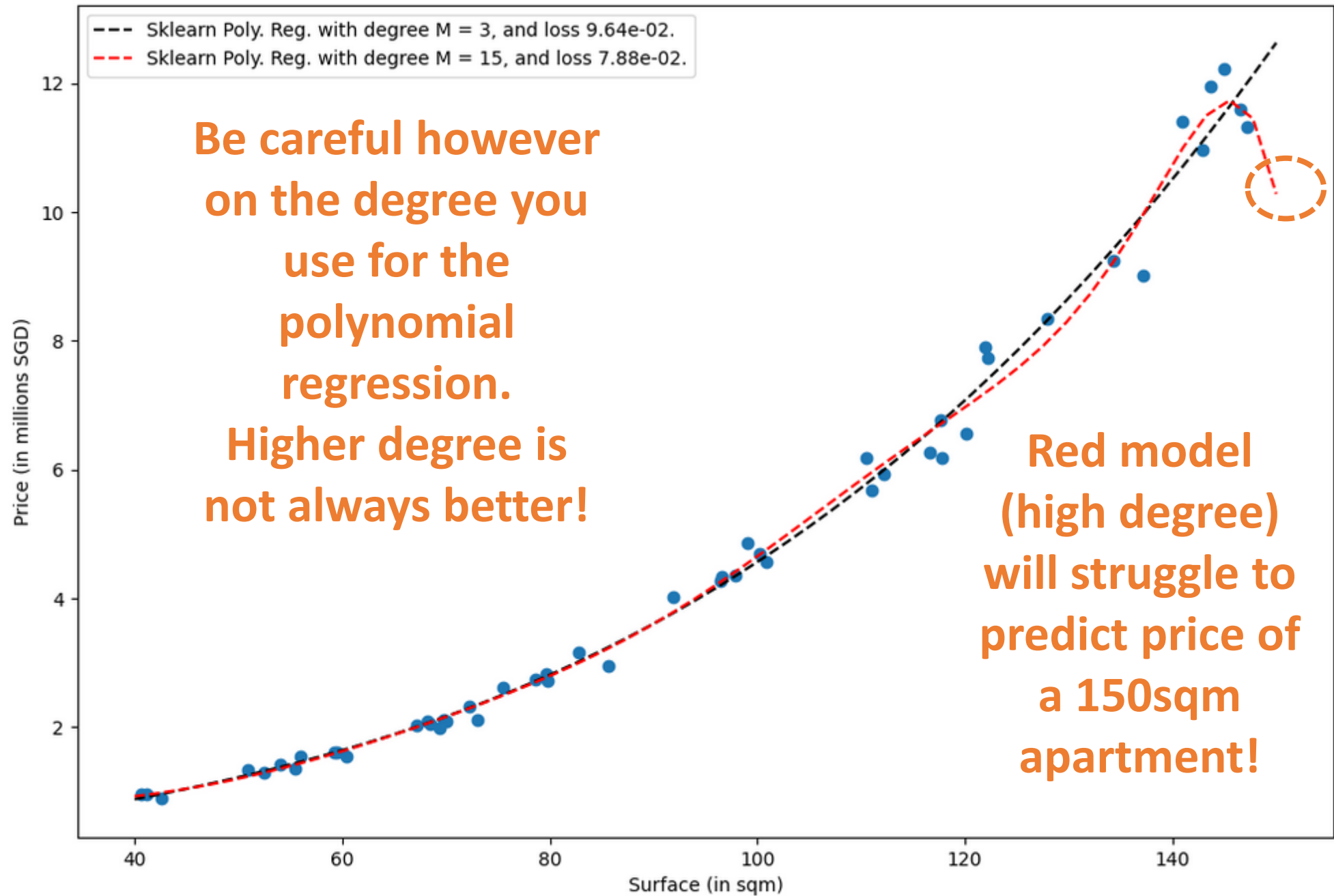
```
[ 2.38010878e-02 -1.30213791e-04  3.58102988e-06] -0.09785310239196843
```

Restricted



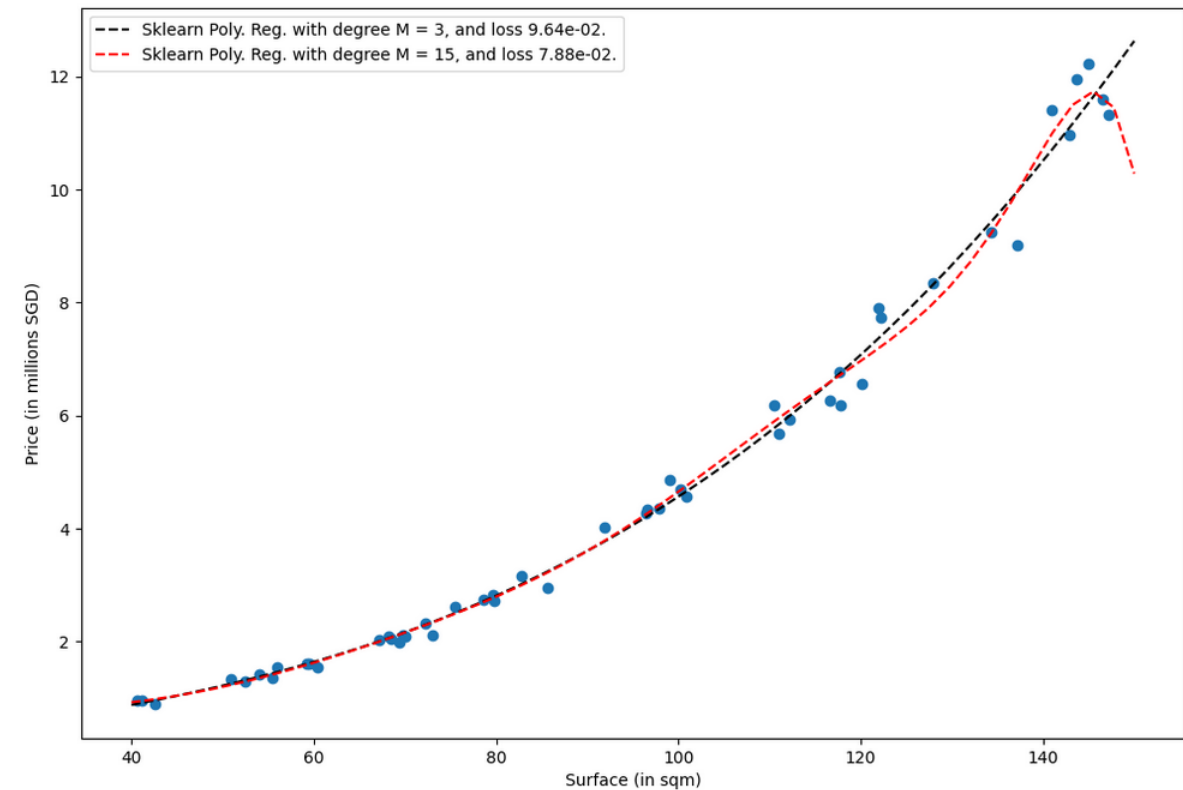
Restricted





Overfitting vs. degree of polynomial

- The red curve shows what happens when using a degree K that is too high compared to the relationship connecting inputs and outputs (it was degree 3 but we use a polynomial regressor with degree 6, which is much higher).
- This phenomenon is called **overfitting**.

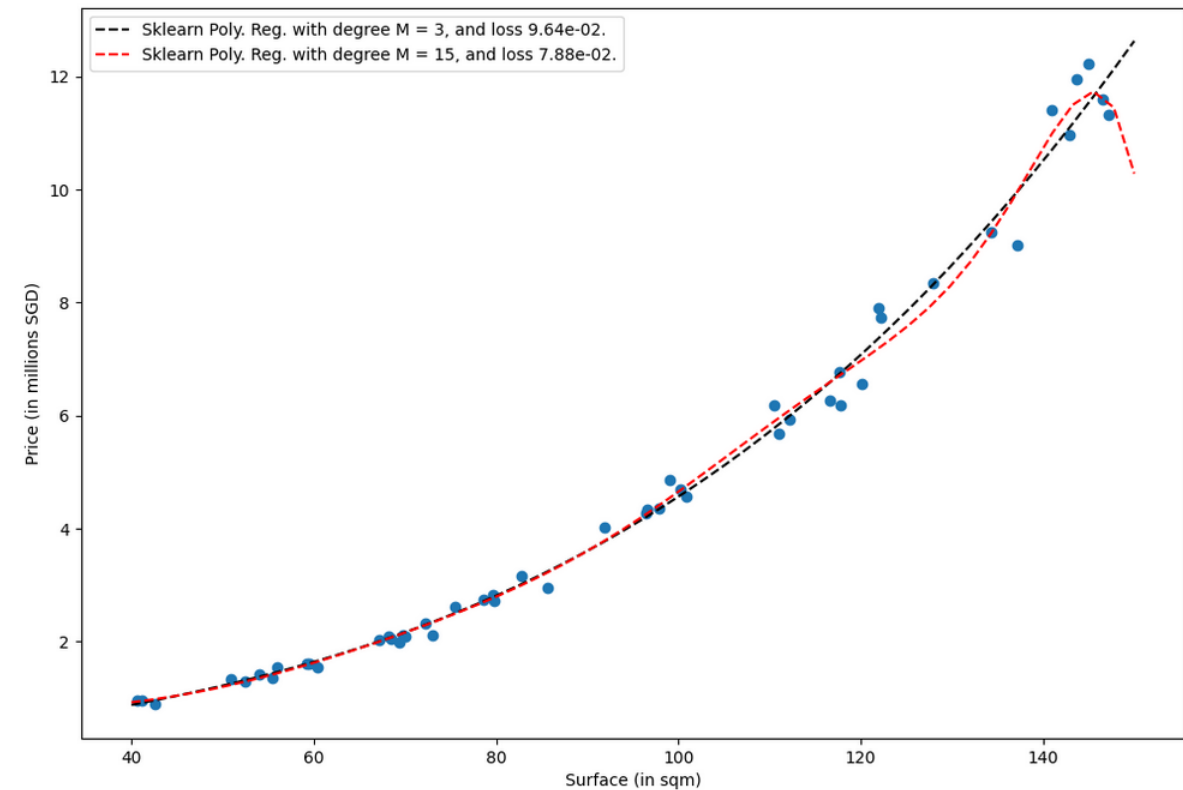


Overfitting vs. degree of polynomial

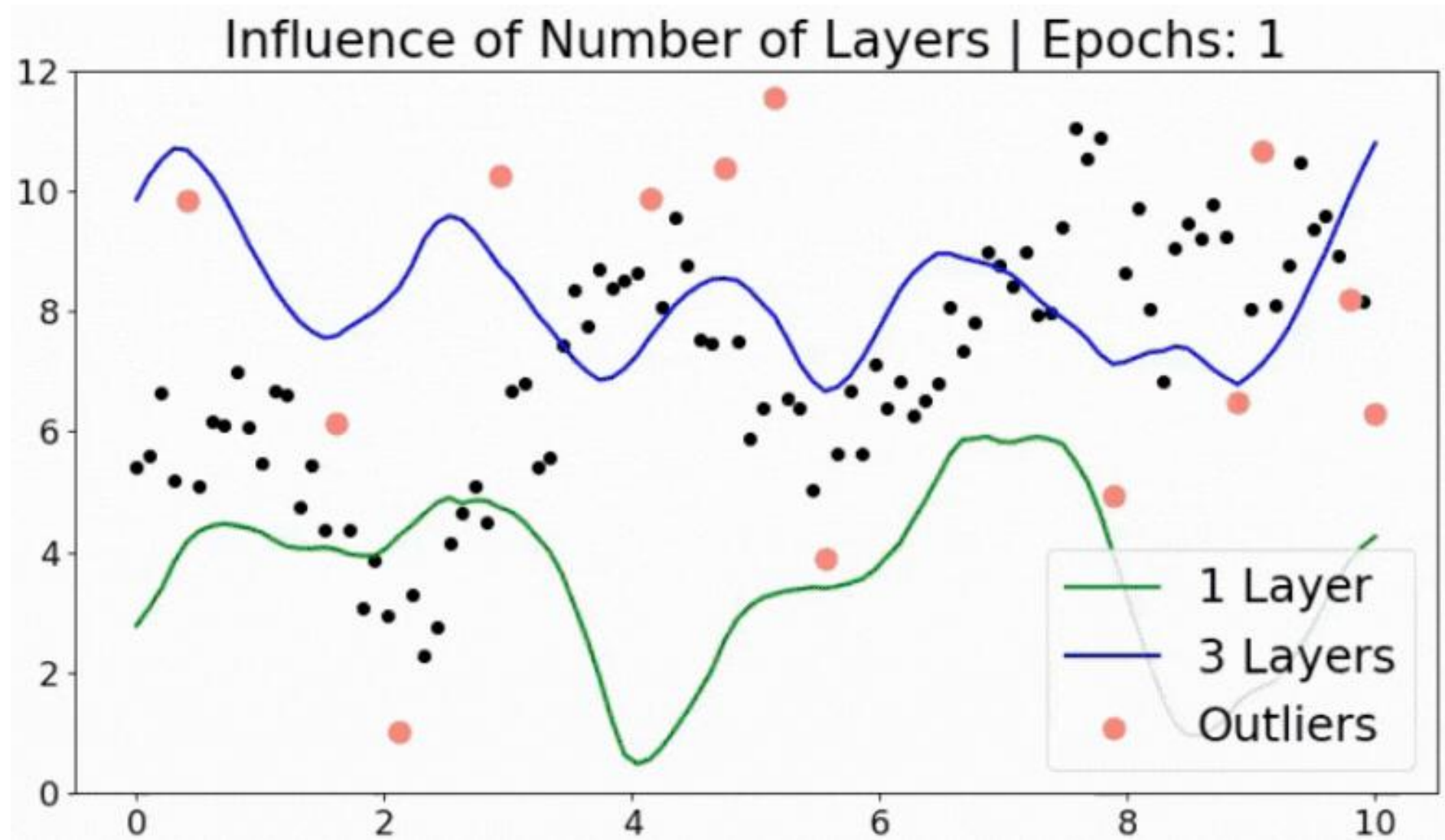
Definition (**Overfitting**):

Overfitting is a phenomenon that occurs in machine learning when a model is trained too well on the training data.

As a result, it may perform poorly on new, unseen data (here an apartment with 150sqm surface, which was not present in the training dataset).



In practice, it may also look like the blue curve, which produces “spikes” to reach outliers. This will lead to questionable predictions on certain input values.



Note: This overfitting (above) was observed on a Neural Network model, not a polynomial regression, but the observation still holds.

Overfitting vs. degree of polynomial

Definition (**Overfitting**):

Overfitting is a phenomenon that occurs in machine learning when a model is trained too well on the training data.

As a result, it may perform poorly on new, unseen data (here an apartment with 150sqm surface, which was not present in the training dataset).

This typically happens when

- a model is trained using too many features in the dataset,
- or has imbalance in the data.

It becomes too complex for the task at hand, resulting in poor **generalization** to new data.

In other words, it memorizes the noise in the training data rather than learning the correct underlying pattern.

Overfitting vs. degree of polynomial

Definition (**Overfitting**):

Overfitting is a phenomenon that occurs in machine learning when a model is trained too well on the training data.

As a result, it may perform poorly on new, unseen data (here an apartment with 150sqm surface, which was not present in the training dataset).

More importantly: It also typically happens **if the model complexity is too high compared to the dataset complexity.**

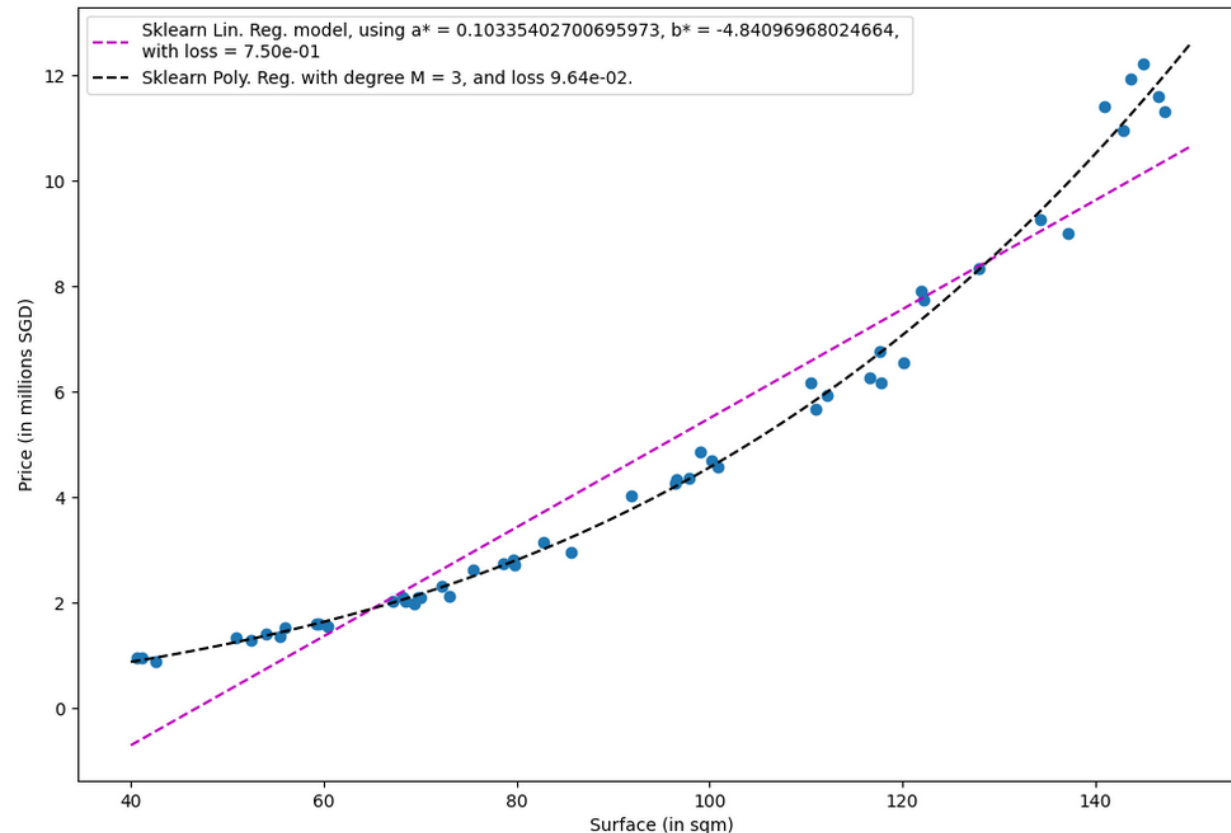
Typically, the model complexity here is the polynomial degree we used for the polynomial regressor: the higher the polynomial degree, the higher the complexity of the model (and its number of trainable parameters)

Underfitting vs. degree of polynomial

Definition (**Underfitting**):

Underfitting is a phenomenon that occurs in machine learning when a model is not capable enough to capture the underlying pattern of the data.

For instance, here, Linear Regression (which is Polynomial Regression with $K = 1$) could not fit the data well.



Underfitting vs. degree of polynomial

Definition (**Underfitting**):

Underfitting is a phenomenon that occurs in machine learning when a model is not capable enough to capture the underlying pattern of the data.

For instance, here, Linear Regression (which is Polynomial Regression with $K = 1$) could not fit the data well.

As the opposite of overfitting, underfitting will typically happen when a model

- is trained with too few features
- or too little data,
- or when the model is not powerful enough to capture the complexity of the task/dataset.

Generalization

Definition (**Generalization**):

Generalization is considered the "holy grail" or ultimate goal of any machine learning model.

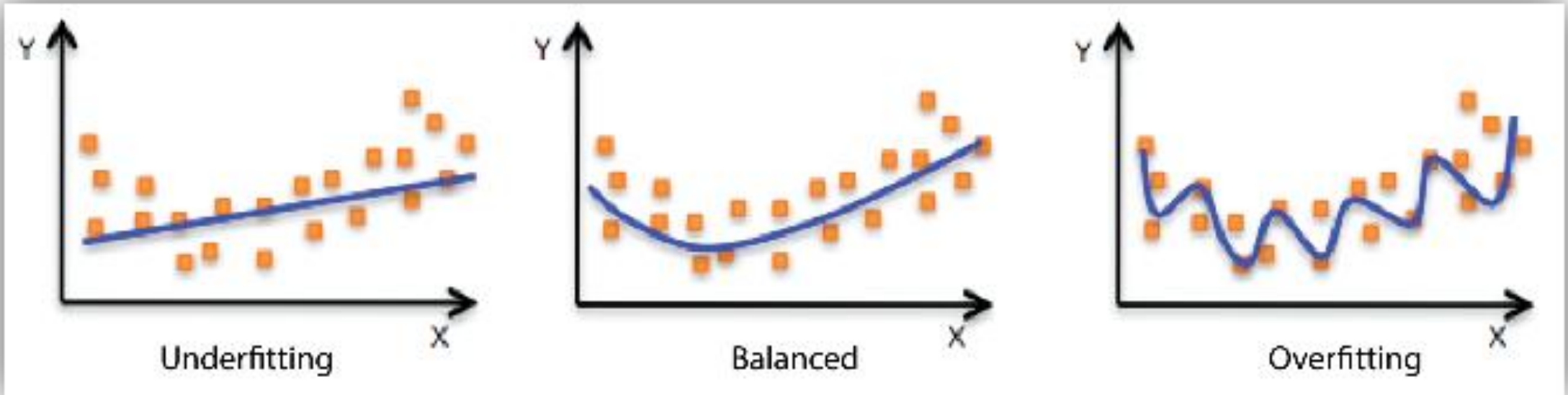
Even though the model was trained on seen data we collected, the purpose of a machine learning model is to **generalize**, i.e. to **make accurate predictions on new, unseen data**.

If a model can make accurate predictions on data it has not seen before, it is said to have **generalized well**, and can then be used in the real world for solving the problem it was designed to solve.

Generalization vs. Overfitting/Underfitting

In general, having a model that is **underfitting** or **overfitting** will lead to poor **generalization**.

We would very much prefer to fit the data “just right”.



While we are at it...

In general, we want to:

- Train the model on the data we collected for this task,
- Confirm that the model can generalize and make accurate predictions on new, unseen data, after training.

Problem: We would prefer to test the model before releasing it in the wild (what if it is wrong?)

Problem #2: Testing its generalization capabilities on the same data that was used for training would NOT confirm generalization.

Solution: Use some of our seen data to train the model, and some of the data (that has not been used for training and is therefore unseen to the model yet) for evaluating its generalization.

Train and test split

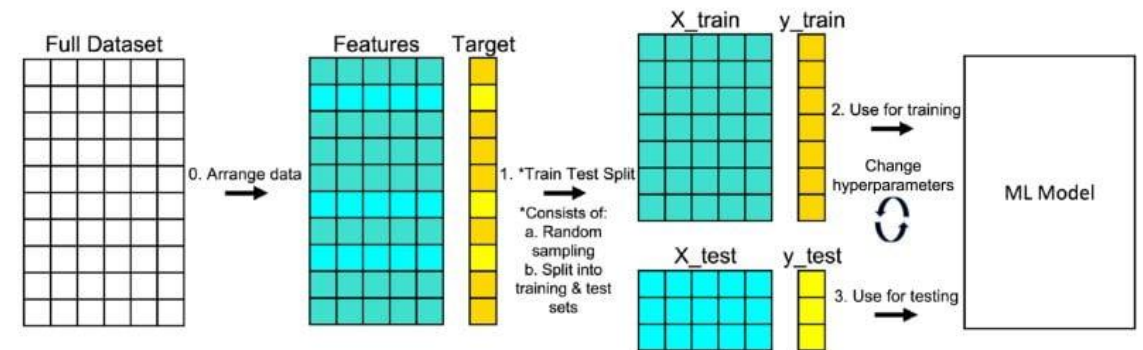
Definition (**Train and Test Split**):

Train and test split is a method used in machine learning to divide a dataset into two subsets:

- The **training dataset** is used to train the model.
- The **test dataset** is used to evaluate the model performance after training.

The idea is to use a portion of the dataset for training and a separate portion for testing so that the model can be evaluated on unseen data.

The typical split is to use 80% of the data for training and 20% for testing. This can however vary depending on the specific use case and the size of the dataset.



Train and test split

```

1 # 80% of the samples will be used for training,
2 # and the remaining 20% will be used to evaluate generalization/overfitting.
3 ratio_train = 0.8
4 split_index = int(n_points*ratio_train)
5 # Training inputs and outputs
6 train_inputs = inputs[:split_index]
7 train_outputs = outputs[:split_index]
8 # Testing inputs and outputs
9 test_inputs = inputs[split_index:]
10 test_outputs = outputs[split_index:]
11 # Display
12 print(train_inputs)
13 print(train_outputs)
14 print(test_inputs)
15 print(test_outputs)

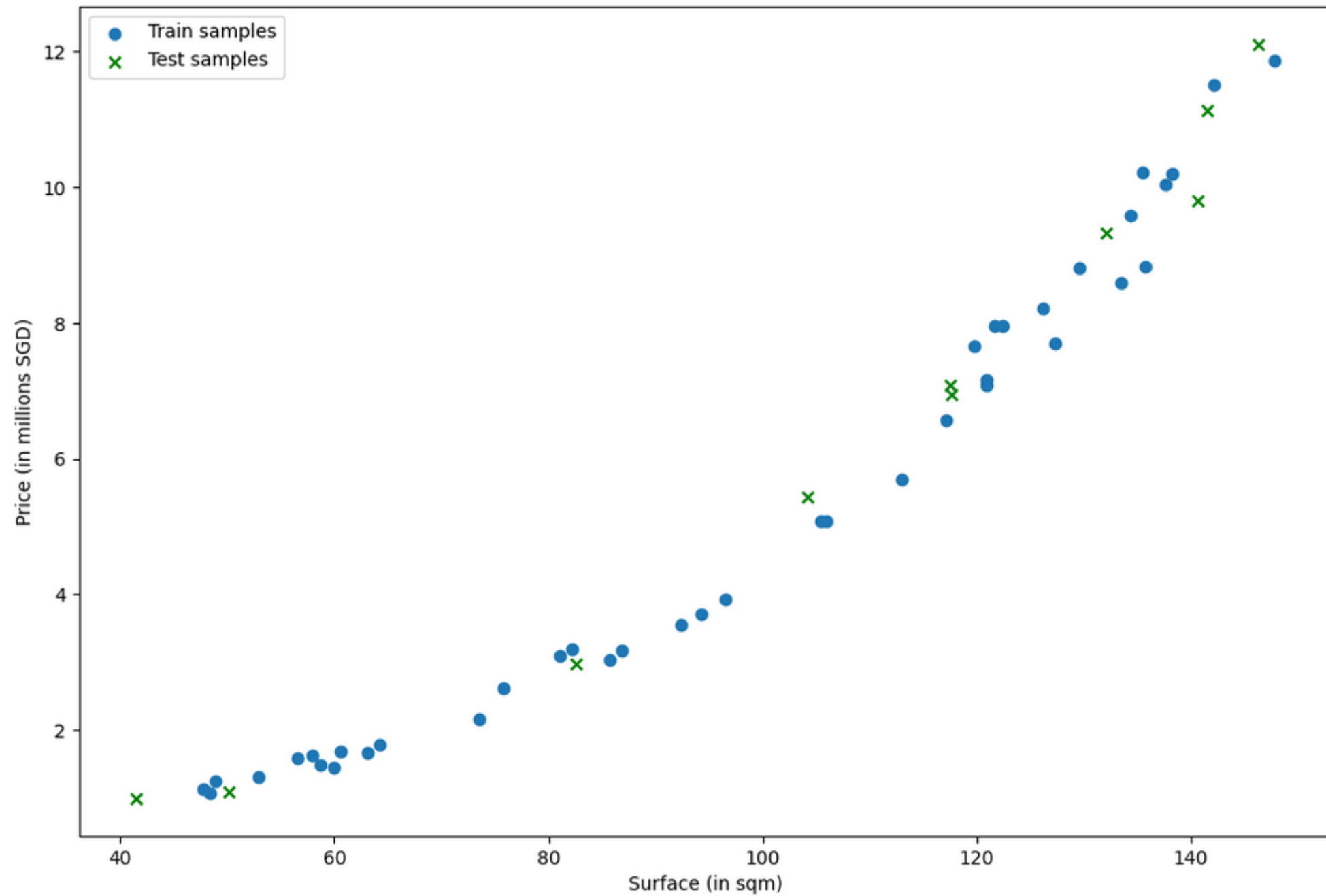
```

```

[ 86.83 129.6 120.89 135.48 82.17 147.74 138.25 63.07 121.6 112.95
 137.55 134.38 122.42 135.72 60.54 75.81 81.02 127.31 56.62 58.69
 48.93 73.57 126.16 57.92 47.77 117.12 59.91 105.88 85.68 96.49
 64.27 119.81 133.44 142.18 120.95 92.42 94.22 105.4 48.36 52.92]
[ 3.171633 8.805667 7.166722 10.224944 3.195151 11.87205 10.206911
 1.676214 7.954078 5.696169 10.049157 9.588631 7.968783 8.827882
 1.693317 2.6239 3.094831 7.700582 1.597057 1.489182 1.247223
 2.166055 8.212138 1.627403 1.12722 6.560241 1.450404 5.072051
 3.031968 3.938955 1.782147 7.652644 8.58658 11.518957 7.088832
 3.563134 3.702923 5.087156 1.071143 1.308982]
[146.31 104.17 50.17 41.5 132.06 140.63 117.5 82.57 117.63 141.56]
[12.111945 5.438864 1.088438 0.998857 9.333922 9.801835 7.095279
 2.982004 6.953392 11.129658]

```

Restricted



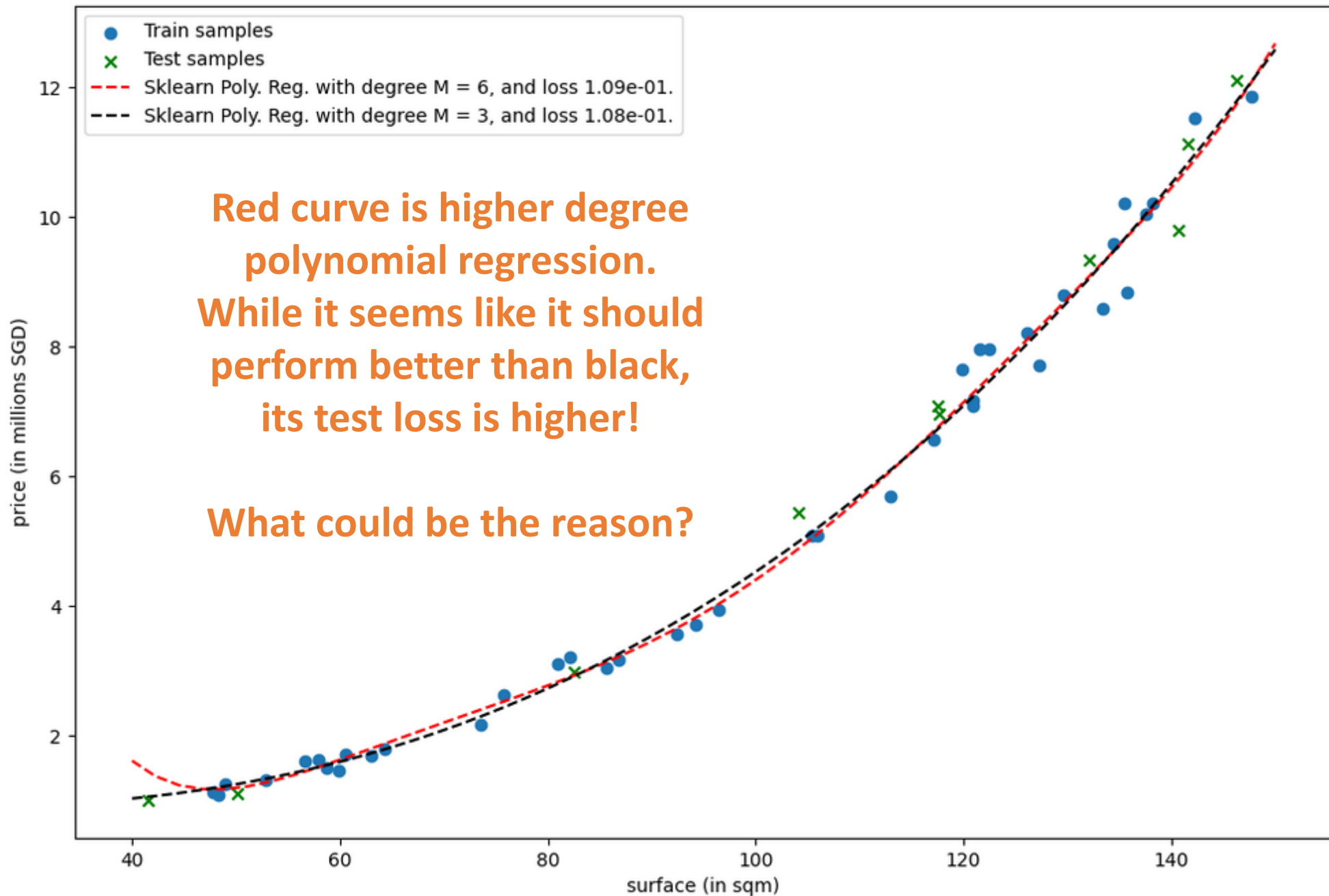
Restricted

Training the Polynomial Regressor, again

- This time, we only train the polynomial regressor on the training data.
- After training, evaluate loss using the test data.

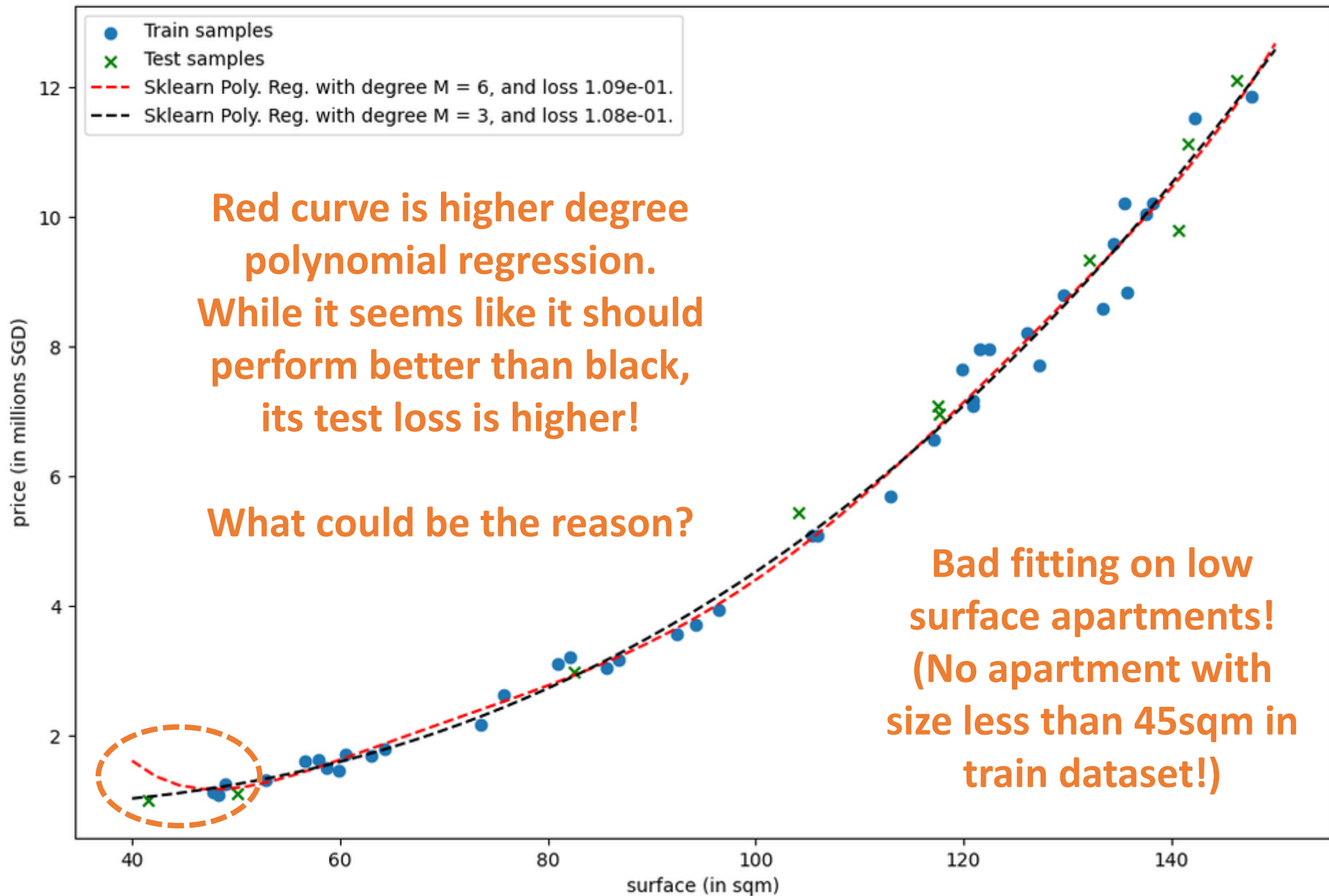
```
1 # Training a Polynomial Regressor
2 n_degree = 6
3 sk_inputs = np.array(train_inputs).reshape(-1, 1)
4 sk_outputs = np.array(train_outputs)
5 sk_poly = PolynomialFeatures(degree = n_degree, include_bias = False)
6 sk_poly_inputs = sk_poly.fit_transform(sk_inputs.reshape(-1, 1))
7 poly_reg_model = LinearRegression()
8 poly_reg_model.fit(sk_poly_inputs, sk_outputs)
9 a_sk = poly_reg_model.coef_
10 b_sk = poly_reg_model.intercept_
11 print(a_sk, b_sk)
```

```
[-5.31103644e+00  1.55941662e-01 -2.34706297e-03  1.92370550e-05
 -8.12548986e-08  1.38734052e-10] 73.25495081585393
```



Red curve is higher degree polynomial regression. While it seems like it should perform better than black, its test loss is higher!

What could be the reason?



Spotting over/underfitting

A typical way to identify overfitting or underfitting is to look at two loss values: the loss calculated using samples from the training set and the one calculated using samples from the test set.

Observation: In general, the train loss has a smaller value than the test one, but they should remain on similar orders of magnitude.

If not, then there might be underfitting or overfitting.

```
1 loss_train = loss_mse_poly(poly_reg_model, n_degree, train_inputs, train_outputs)
2 print(loss_train)
3 loss_test = loss_mse_poly(poly_reg_model, n_degree, test_inputs, test_outputs)
4 print(loss_test)
```

5.46e-02

7.99e-01

Generalization vs. Train/Test Distributions

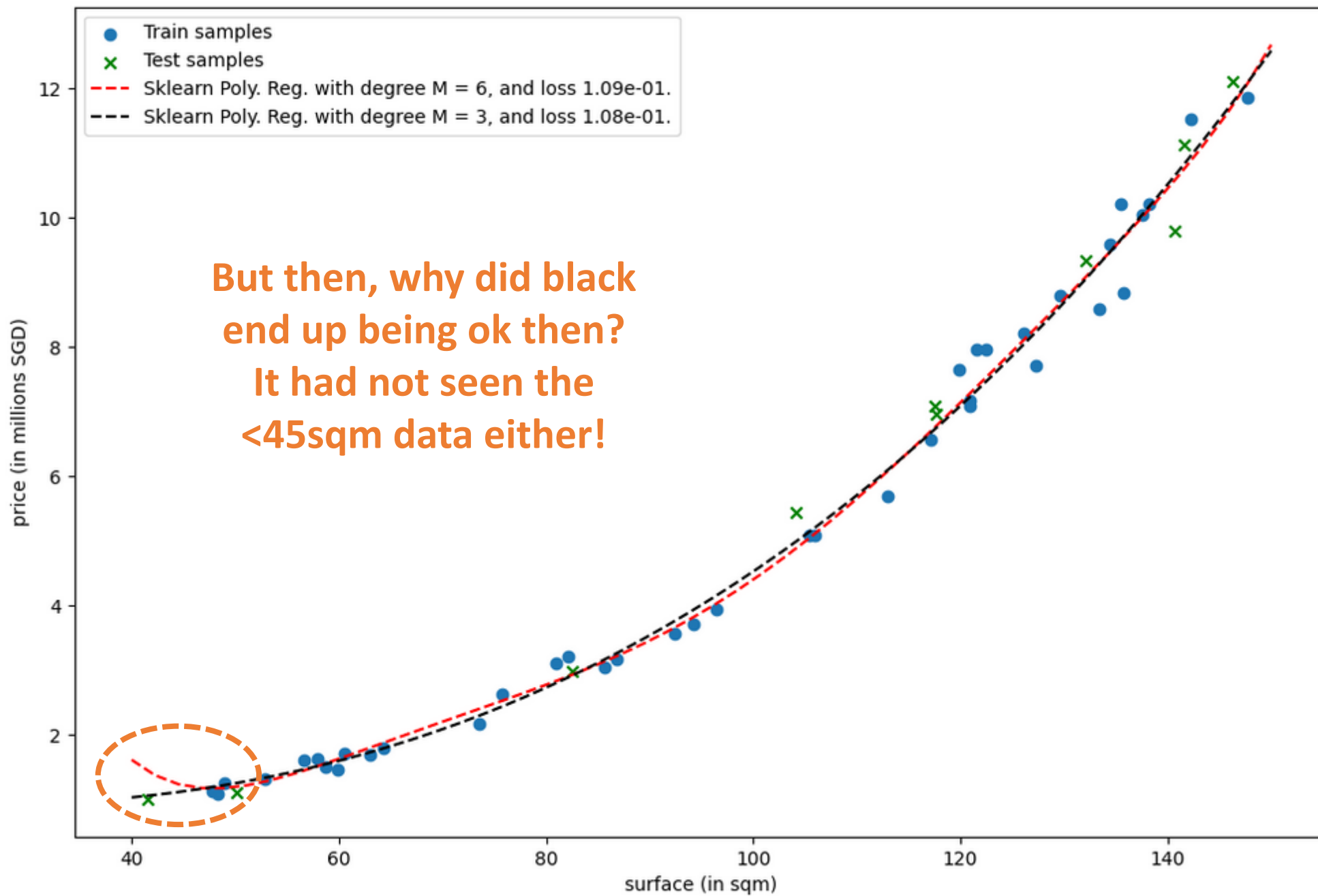
Definition (The need for train and test distribution similarity):

In order to generalize well, it is important that the **train and test set are following similar distributions**. Mathematically speaking, we want:

$$P_{train} \approx P_{test}$$

Here, we ended up having a problem, as our training dataset did not contain any apartments with size lower than 45sqm.

I mean, you would not train a model on Singaporean apartments and test it by predicting the price of apartments in Kuala Lumpur, right?



Regularization

Definition (**Regularization**):

Regularization consists of techniques, whose purpose is to help the model to avoid overfitting.

It can typically be done by making sure the model is not too complex for the data, but many other approaches exist.

In general,

- **Scale model and dataset complexity:** the more complex the data is (in terms of features), the more complex the model can/should be.
- **Scale model and dataset size:** the more data you have (in terms of quantity of individual samples in dataset), the more complex the model can be.

Regularization

Definition (**Regularization**):

Regularization consists of techniques, whose purpose is to help the model to avoid overfitting.

It can typically be done by making sure the model is not too complex for the data, but many other approaches exist.

A common approach to regularize a model consists of **adding a penalty term to the loss function**.

The penalty term, also known as a **regularization term**, discourages the model from assigning too much weight to any one feature, i.e. making a certain parameter a_k too prominent (and that is often the reason leading to overfitting).

Regularization

Definition (**Ridge Regression**):

The **Ridge Regressor** is a polynomial regressor, like described earlier.

Its only difference is that **the loss function includes an additional regularization term $\lambda R(a, b)$** .

This regularization term consists of the **sum of the squared values for the trainable parameters**, i.e.

$$\lambda R(a, b) = \frac{\lambda}{2N} \left(\sum_k^M (a_k)^2 + (b)^2 \right)$$

Why does it work?

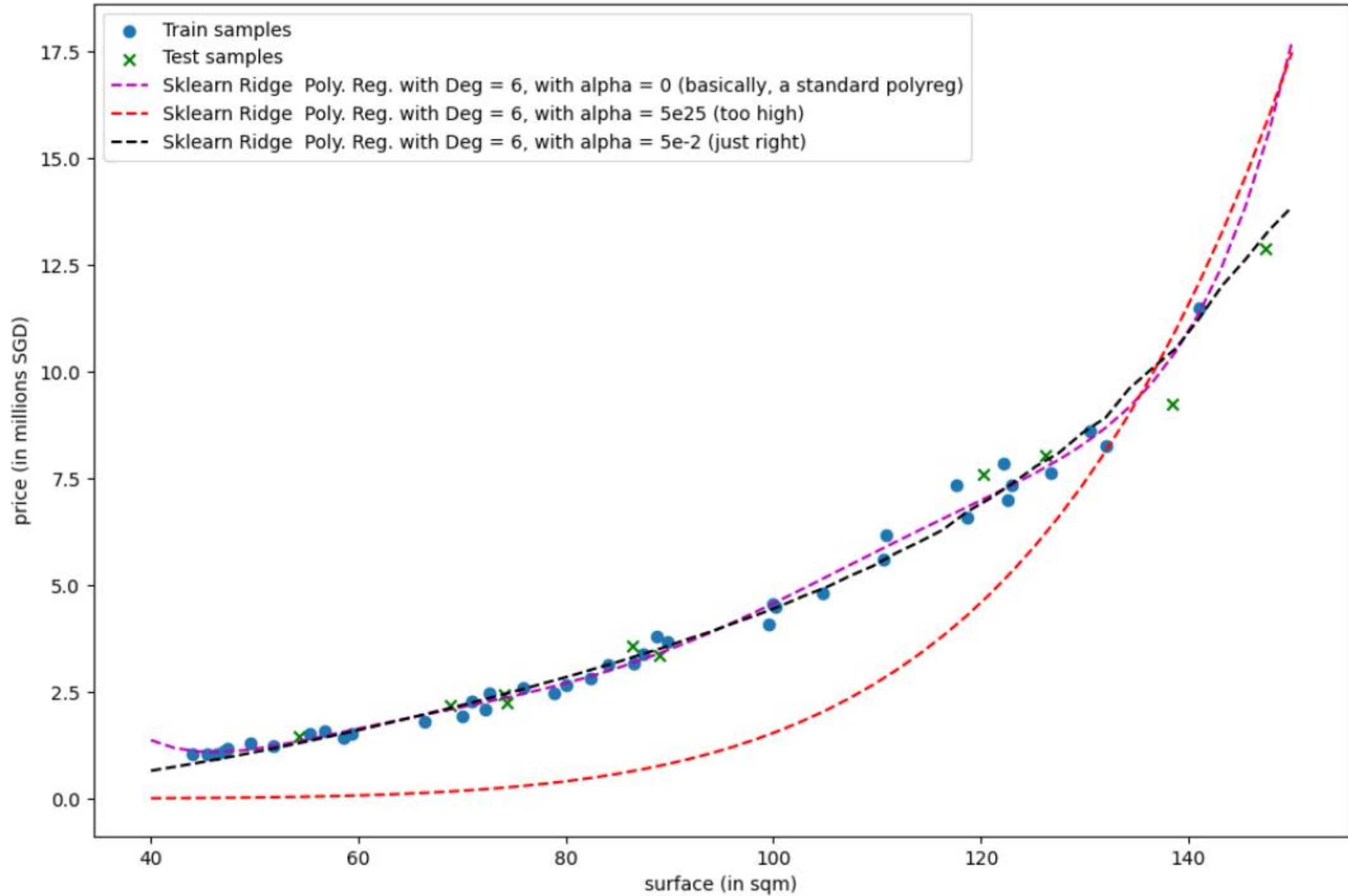
- This **regularization term encourages the model to assign low values to the model parameters**.
- This will, in turn, lead to less overfitting.
- Indeed, overfitting often occurs when high values are assigned to coefficients for high degrees (i.e. coefficients a_k with high k values).

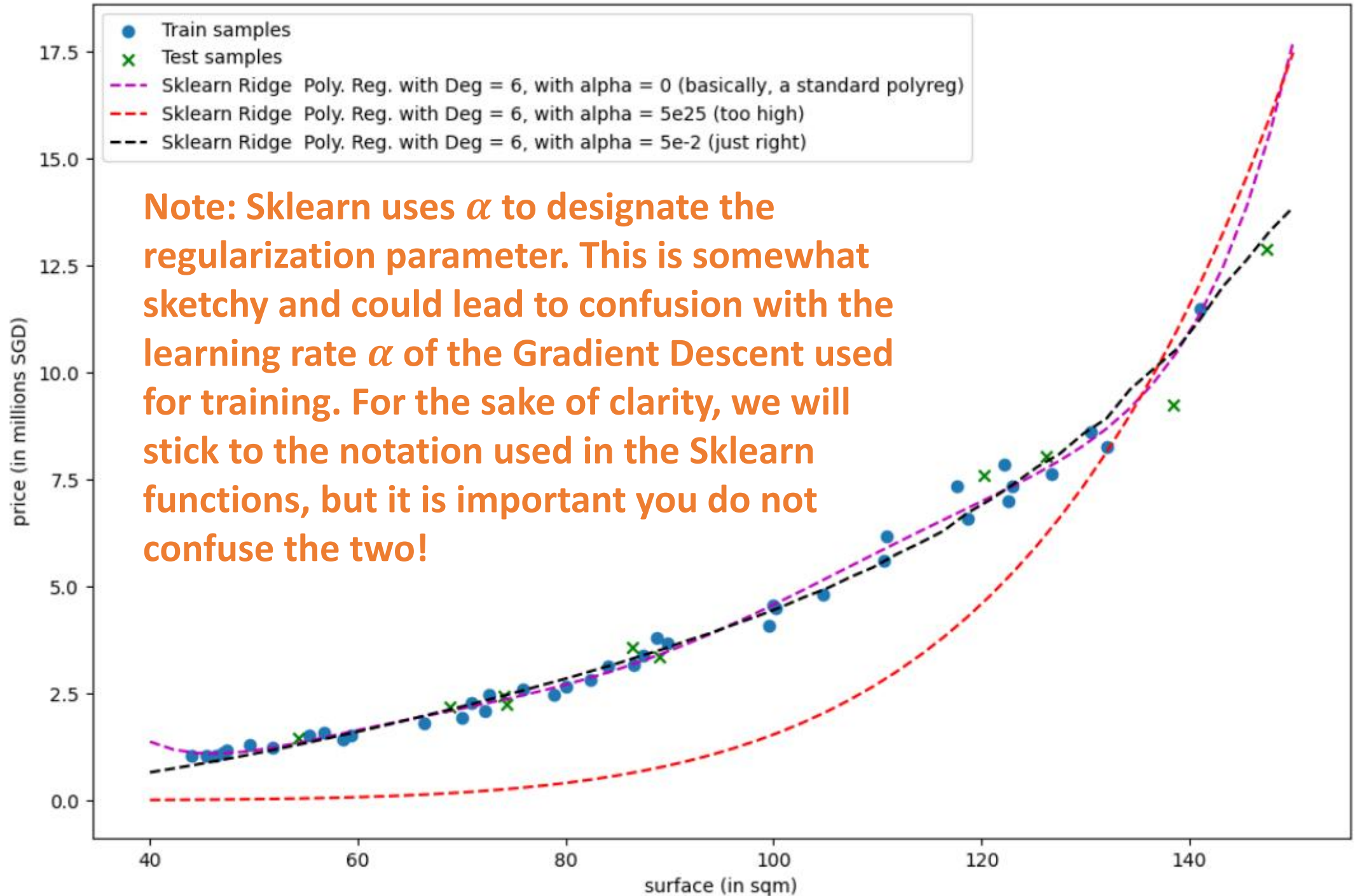
Regularization and hyperparameter λ

Just like the learning rate α in the GD algorithm, the parameter λ must be **manually decided** and is used to weight the importance of the regularization term compared to the MSE.

- If the **value for λ is small**, the loss function will consist **mostly of MSE** and our model will likely suffer from the same **overfitting** problems we had earlier.
- If the **value of λ is too high** however, the model will **mostly ignore the MSE** part of the loss function, which will lead to a model not fitting (or **underfitting**) the data.

We say that λ is a **hyperparameter**, and choosing the right, balanced value to use is commonly referred to as **hyperparameter tuning**.





Regularization with other things than L2?

Also possible to use the sum of **absolute** values (or what we call a L1 regularization), making a **Lasso Regressor** instead of a **Ridge Regressor**.

Why use L1 regularization over L2 regularization?

- L1 regularization results in some weights being exactly equal to zero, which can be used for feature selection.
- L2 regularization will not produce sparse models, and it will only shrink the weights.

Why prefer L1 over L2 (and vice versa)? Have a look at this, if curious!

<https://medium.com/@fernando.dijkinga/explaining-l1-and-l2-regularization-in-machine-learning-2356ee91c8e3>

Regularization with other things than L2?

Also possible to use the sum of absolute values **AND** squared values as regularization, i.e. combining the **L1** and **L2** regularizations together.

Essentially getting the best of both worlds, and making an **Elastic Regressor** instead! (MSE + L1 Reg + L2 Reg)

$$L_{ER} = \underbrace{\frac{1}{N} \sum_{i=1}^N (p(x_i) - y_i)^2}_{\text{MSE}} + \underbrace{\lambda_1 \left(\sum_{k=1}^K |a_k| + |b| \right)}_{\text{L1}} + \underbrace{\lambda_2 \left(\sum_{k=1}^K (a_k)^2 + b^2 \right)}_{\text{L2}}$$

Done for today!

Let us know if you had any installation issues (PyTorch + CUDA) and need help.
Otherwise, feel free to play with the notebooks.

Quick parenthesis: The hyperplane concept

Definition (Hyperplanes in Linear Algebra):

A **hyperplane** is a subspace of one dimension less than the ambient space.

In the case of linear regression with one input feature and one output feature, the ambient space is of dimension 2 (that is 1+1).

The Linear Regression model equation, is defined as

$$y = ax + b$$

This is the equation of a line, which is a 1D subspace, and therefore a hyperplane of the 2D ambient space.

The same happens with the polynomial regression, but at a higher degree instead of just 2D/1D.